

Elvin Esguerra

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Education

The University of Missouri–Columbia, Columbia, MO
Bachelor of Science in Mechanical Engineering

Expected May 2029
GPA: 4.00/4.00

Experience

Summer Manufacturing Group Leader, *General Motors*

Jun. 2026 – Aug. 2026

- Directed 36 operators across 3 stamping presses to exceed 3,700 parts daily while meeting quality and safety requirements.
- Eliminated 40 minutes of recurring weekly downtime by developing 2 standardized robot-programming work instructions, adding capacity for approximately 200 parts per week.
- Performed root cause analysis on stamping defects and press/robot faults, coordinating controls, maintenance, and production teams to isolate failures and restore production.

Publications

Quantifying Convective Aerodynamic Heating During Ascent

Jan. 2026 – Mar. 2026

Published and Presented at AIAA Region V Conference | <https://arc.aiaa.org/doi/epdf/10.2514/6.2026-111868>

- Developed a compressible 2D axisymmetric ANSYS Fluent model using SST $k-\omega$ and targeted near-wall refinement to resolve stagnation-region and boundary-layer heating across 5 ascent conditions.
- Applied flight-derived Mach, pressure, and temperature conditions to quantify peak stagnation-region heat flux of 2.46 MW/m^2 at Max-Q.
- Benchmarked CFD against Sutton–Graves and a 3-grid mesh study, finding 183%–452% higher stagnation heat flux and continued nose-tip mesh sensitivity.

Engineering Projects

Argonia Cup Project Manager, *Mizzou Space Program*

May 2026 – Present

- Leading a 30+ member multidisciplinary team through design, manufacturing, and testing of a competition rocket targeting a 7,000-ft apogee with active air-brake development.
- Streamlined fin aerodynamic optimization by developing a Python wrapper for OpenRocket to parameterize geometry and automate simulation sweeps, reducing analysis from 2 weeks to 4 hours.
- Characterized flight-condition uncertainty through 250 Monte Carlo simulations varying atmospheric and launch conditions, with 99.2% meeting second-stage ignition criteria.

IREC Nose Cone Aerodynamic Analysis, *Mizzou Space Program*

Apr. 2026 – May 2026

- Simulated Mach 2.12 compressible supersonic flow in ANSYS Fluent using OpenRocket-derived Max-Q conditions and a 12-in aft extension to shift base separation downstream.
- Generated a 474k-cell structured mesh with 20 inflation layers, a Body of Influence, upstream/downstream refinement, and average y^+ of 1.6 using SST $k-\omega$.
- Calculated 245.5 N of total forebody drag with a reported drag coefficient of 0.123, using an independent MATLAB model based on governing equations to support the predicted pressure-drag magnitude.

Undergraduate Research Assistant, *Semiconductor Lab*

Jan. 2026 – May 2026

- Designed a 3-point bending system in SolidWorks for $60 \text{ mm} \times 60 \text{ mm}$ mechanoluminescent samples using stepper-driven loading and 3D-printed components for under \$40.
- Wired and programmed an Arduino controller to command stepper displacement and automate user-defined loading cycles for mechanoluminescent-material testing.

IREC Payload, *Mizzou Space Program*

Aug. 2025 – May 2026

- Designed and fabricated a 5.75-lb modular payload around 303 stainless-steel rails, G10 fiberglass disks, and 3D-printed enclosures for a 20 ft/s landing requirement.
- Secured telemetry, batteries, Arduino/IMU electronics, Raspberry Pi 5, camera, and antenna hardware while maintaining structural load paths, RF access, and serviceability.
- Drafted 8 GD&T drawings using datum references and positional tolerances to control critical mounting features through fabrication and assembly.

Technical Skills

Simulation and Analysis: ANSYS Fluent, ANSYS Mechanical, SolidWorks Simulation, CFD, FEA, OpenRocket

CAD and Manufacturing: SolidWorks, AutoCAD, Rapid Prototyping, 3D Printing, Design for Manufacturing, GD&T

Programming: Python, MATLAB, SQL

Certifications: CSWA, CSWA–Simulation